



Figure 3: Applications of AR, VR, and MR across industries

innovate marketing. For instance:

- AR is used for virtual shopping.
- VR can create immersive virtual stores where customers can browse and shop in a digital environment.
- MR can enhance in-store navigation by overlaying digital maps and product information in the physical retail space.

Oil and gas industry: In the oil and gas sectors, AR, VR, and MR technologies can be used to improve various aspects of operations, safety, training, and maintenance.

- AR can be used for maintenance and inspection of oil rigs and equipment, providing technicians with real-time data and instructions.
- VR can be utilised for virtual training simulations, allowing workers to practice emergency responses in a secure environment.
- MR can be used for remote monitoring of oil field operations, overlaying digital information onto the physical environment for improved decision-making.

Exploring simulation tools and devices for AR, VR, and MR applications

Simulation tools are instrumental in creating realistic virtual environments for training, education, and entertainment purposes. These tools enable developers to design and simulate complex scenarios. Moreover, the choice of devices for AR, VR, and MR applications significantly impacts the quality and accessibility of these experiences. From high-end VR headsets to smartphone-based AR applications, the diverse range of devices available caters to various needs and preferences. Understanding

the capabilities of simulation tools and selecting the right devices are essential steps in harnessing the full potential of AR, VR, and MR technologies for innovative and impactful applications.

Tables 2 and 3 provide a quick overview of popular simulation tools and the devices used for the development of immersive applications.

With the advent of AR/VR/MR technologies, the world is progressively transitioning towards an environment where the real and virtual realms can coexist in harmony. This trend is being accelerated by the widespread adoption of the cloud, edge, and various machine learning-based applications. **END** 🐧

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